

Solving Linear Equations by Crout's Method

Omkar Kashid

M.Sc. Mathematics

Problem

Solve the following system of linear equations by Crout's method:

$$4x_1 + x_2 + x_3 = 4,$$

$$x_1 + 4x_2 - 2x_3 = 4,$$

$$3x_1 + 2x_2 - 4x_3 = 6.$$

Matrix Form

The given system can be written in the form

$$AX = B$$

where

$$A = \begin{bmatrix} 4 & 1 & 1 \\ 1 & 4 & -2 \\ 3 & 2 & -4 \end{bmatrix}, \quad X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad B = \begin{bmatrix} 4 \\ 4 \\ 6 \end{bmatrix}.$$

Crout's Method

In Crout's method, the coefficient matrix A is factorized as

$$A = LU.$$

Here, L is a lower triangular matrix and U is an upper triangular matrix with diagonal elements equal to 1.

$$L = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix}$$

and

$$U = \begin{bmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix}.$$

Finding L and U

Since

$$A = LU,$$

we compare the corresponding elements of the matrices.

From the first row,

$$l_{11} = 4$$

$$l_{11}u_{12} = 1 \quad \Rightarrow \quad u_{12} = \frac{1}{4}$$

$$l_{11}u_{13} = 1 \quad \Rightarrow \quad u_{13} = \frac{1}{4}.$$

From the second row,

$$l_{21} = 1$$

$$l_{21}u_{12} + l_{22} = 4$$

$$1 \left(\frac{1}{4} \right) + l_{22} = 4$$

$$l_{22} = \frac{15}{4}.$$

Finding L and U

For the $(2, 3)$ position,

$$l_{21}u_{13} + l_{22}u_{23} = -2.$$

Therefore,

$$1 \left(\frac{1}{4} \right) + \frac{15}{4}u_{23} = -2.$$

Hence,

$$u_{23} = -\frac{3}{5}.$$

Completing the Third Row

For the third row,

$$l_{31} = 3.$$

Also,

$$l_{31}u_{12} + l_{32} = 2.$$

Thus,

$$3\left(\frac{1}{4}\right) + l_{32} = 2,$$

so

$$l_{32} = \frac{5}{4}.$$

Finally,

$$l_{31}u_{13} + l_{32}u_{23} + l_{33} = -4.$$

Hence,

$$l_{33} = -4.$$

Matrix $A = LU$

The coefficient matrix is

$$A = \begin{bmatrix} 4 & 1 & 1 \\ 1 & 4 & -2 \\ 3 & 2 & -4 \end{bmatrix}.$$

Using Crout's method,

$$A = LU$$

Therefore,

$$\begin{bmatrix} 4 & 1 & 1 \\ 1 & 4 & -2 \\ 3 & 2 & -4 \end{bmatrix} = \begin{bmatrix} 4 & 0 & 0 \\ 1 & \frac{15}{4} & 0 \\ 3 & \frac{5}{4} & -4 \end{bmatrix} \begin{bmatrix} 1 & \frac{1}{4} & \frac{1}{4} \\ 0 & 1 & -\frac{3}{5} \\ 0 & 0 & 1 \end{bmatrix}$$

$$A = L \times U$$

Forward Substitution

We have

$$AX = B.$$

Since

$$A = LU,$$

we get

$$LUX = B.$$

Put

$$UX = Z.$$

Therefore,

$$LZ = B.$$

Hence,

$$\begin{bmatrix} 4 & 0 & 0 \\ 1 & \frac{15}{4} & 0 \\ 3 & \frac{5}{4} & -4 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \\ 6 \end{bmatrix}.$$

Solving $LZ = B$

From the first equation,

$$4z_1 = 4$$

Hence,

$$z_1 = 1.$$

From the second equation,

$$z_1 + \frac{15}{4}z_2 = 4.$$

Therefore,

$$1 + \frac{15}{4}z_2 = 4$$

so

$$z_2 = \frac{4}{5}.$$

Solving $LZ = B$

From the third equation,

$$3z_1 + \frac{5}{4}z_2 - 4z_3 = 6.$$

Substituting $z_1 = 1$ and $z_2 = \frac{4}{5}$,

$$3 + \frac{5}{4} \left(\frac{4}{5} \right) - 4z_3 = 6.$$

Hence,

$$z_3 = -\frac{1}{2}.$$

Therefore,

$$Z = \begin{bmatrix} 1 \\ \frac{4}{5} \\ -\frac{1}{2} \end{bmatrix}$$

Backward Substitution

Now solve

$$UX = Z.$$

Therefore,

$$\begin{bmatrix} 1 & \frac{1}{4} & \frac{1}{4} \\ 0 & 1 & -\frac{3}{5} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ \frac{4}{5} \\ -\frac{1}{2} \end{bmatrix}.$$

From the third equation,

$$x_3 = -\frac{1}{2}.$$

Finding x_2 and x_1

From the second equation,

$$x_2 - \frac{3}{5}x_3 = \frac{4}{5}.$$

Substituting $x_3 = -\frac{1}{2}$,

$$x_2 + \frac{3}{10} = \frac{4}{5}.$$

Thus, $\boxed{x_2 = \frac{1}{2}}$. From the first equation,

$$x_1 + \frac{1}{4}x_2 + \frac{1}{4}x_3 = 1.$$

Substituting the values,

$$x_1 + \frac{1}{4} \left(\frac{1}{2} \right) + \frac{1}{4} \left(-\frac{1}{2} \right) = 1.$$

Therefore, $\boxed{x_1 = 1}$.

Hence,

$$x_1 = 1, \quad x_2 = \frac{1}{2}, \quad x_3 = -\frac{1}{2}$$

Therefore, the solution of the given system is

$$(x_1, x_2, x_3) = \left(1, \frac{1}{2}, -\frac{1}{2}\right)$$

Conclusion

- Crout's method factorizes the coefficient matrix as

$$A = LU.$$

- L is a lower triangular matrix.
- U is an upper triangular matrix with diagonal elements equal to 1.
- The system is solved in two steps:

$$LZ = B$$

followed by

$$UX = Z.$$

- The required solution is

$$\left(1, \frac{1}{2}, -\frac{1}{2}\right).$$

Thank You